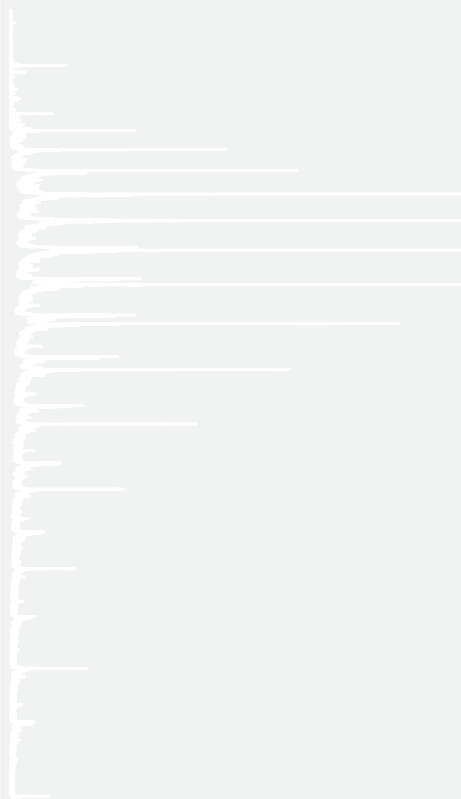




MASS SPECTROMETRY
LABORATORY



JMS-T200 AccuTOF Gcx-plus

Manufacturer: [JEOL USA Inc.](#)
Peabody MA

Features

Ion Sources: Electron Ionization (EI)
Field Desorption (FD)
Field Ionization (FI)
Ion Polarity: Positive and negative
Mass Range: 7-10 000 Da
Resolution: >10 000 mass resolution @ m/z 609
Mass Accuracy: +/- ~1.5 mDa
GC System: Agilent 7890B



AccuTOF 4G DART Results Format

EI-MS Spectrum

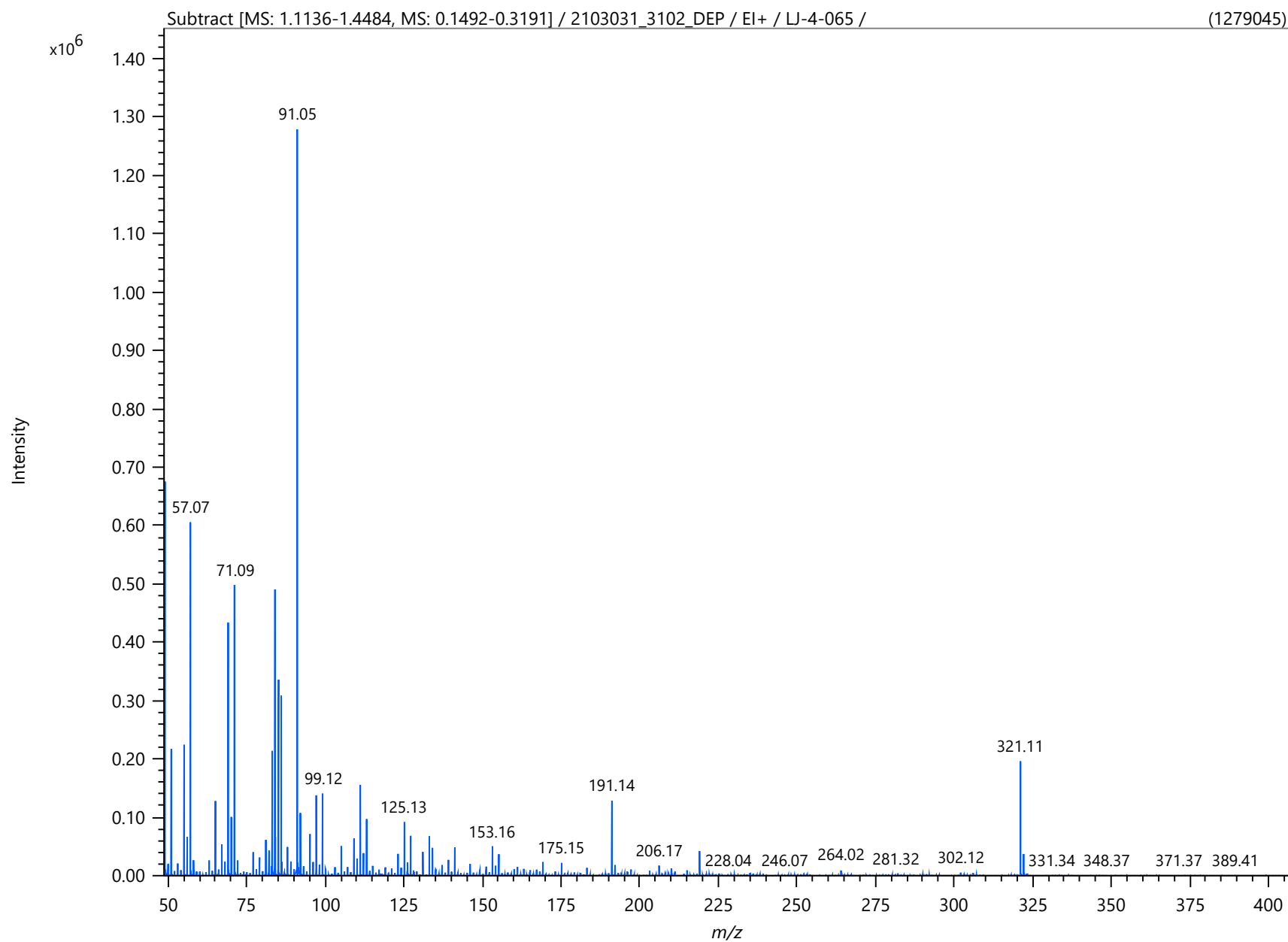
The *msAxel* Spectrum Report consists of a profile mass spectrum showing m/z versus intensity. Spectral peaks are labeled with m/z values rounded to two decimal places. The standard data acquisition range is m/z 50-850 Da but printed mass spectra are normally expanded to display an appropriate region of interest. EI mass spectra are normally acquired in positive ion mode and the analyte species of interest appear as radical molecular ions $[M]^+$.

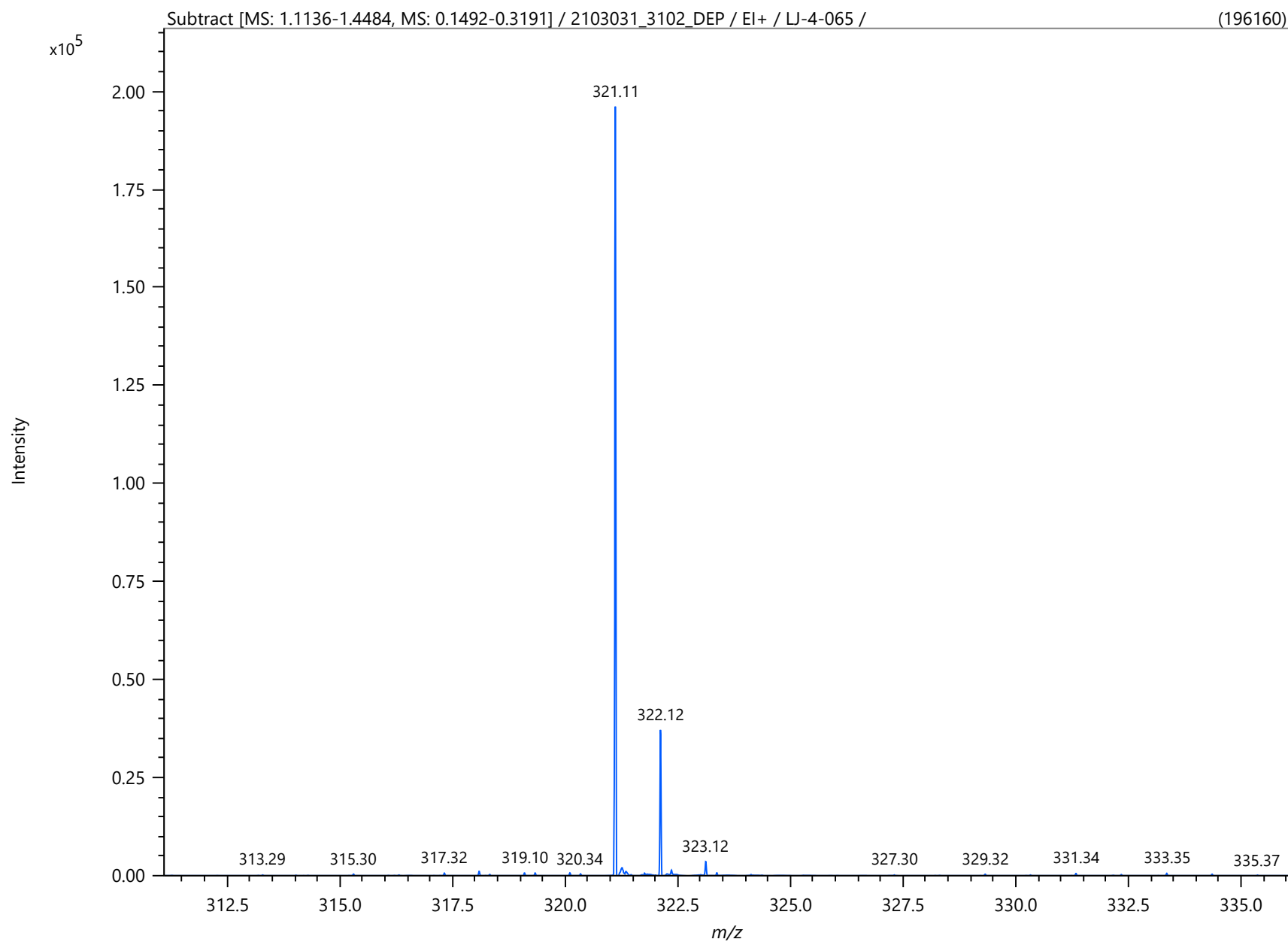
High Resolution Mass Spectrometry (HRMS) Report

The EI-HRMS report is generated using the *Elemental Composition* feature implemented in the JEOL *msAxel* software package. The report provides the measured m/z value and a list of possible molecular formulae and calculated exact m/z values. All of the values contained in the report correspond to the ionic species of interest, despite the table heading. All ionic formulae are reported as charged species accounting appropriately for the mass of the electron (0.00054 Da).

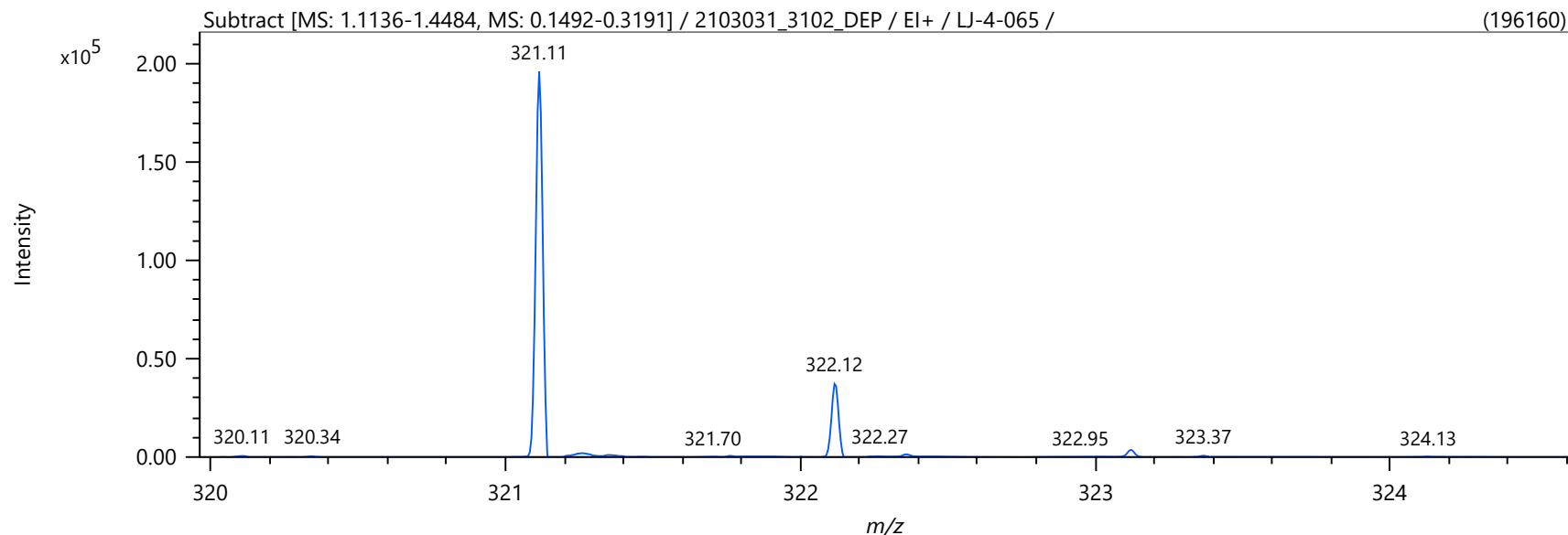
Suggested reporting format for compound characterization:

"HRMS (EI-TOF+): m/z $[M]^+$, calc'd for $C_{18}H_{15}NF_4^+$ m/z 321.11351, measured m/z 321.1142"





Spectrum



Elemental Composition

Parameters

Tolerance: ± 5.00 mDa
 Electron: Odd/Even
 Charge: +1
 DBE: -1.5 - 40.0

Elements Set 1:

Symbol	C	H	N	O	F
Min	0	0	0	0	4
Max	50	100	10	10	4

Results

Mass	Intensity	Formula	Calculated Mass	Mass Difference [mDa]	Mass Difference [ppm]	DBE
321.11416	196160.25	C ₁₈ H ₁₅ N ₄ F ₄	321.11351	0.65	2.03	10.0
		C ₅ H ₁₃ N ₁₀ O ₂ F ₄	321.11536	-1.19	-3.72	2.5
		C ₇ H ₁₅ N ₇ O ₃ F ₄	321.11670	-2.54	-7.90	2.0
		C ₁₅ H ₁₇ O ₃ F ₄	321.11083	3.33	10.37	5.5
		C ₉ H ₁₇ N ₄ O ₄ F ₄	321.11804	-3.88	-12.08	1.5
		C ₁₃ H ₁₅ N ₃ O ₂ F ₄	321.10949	4.67	14.55	6.0